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Three-stage integrated co-digestion of pig manure and rice straw combined with biogas slurry reflux: Performance and application evaluation

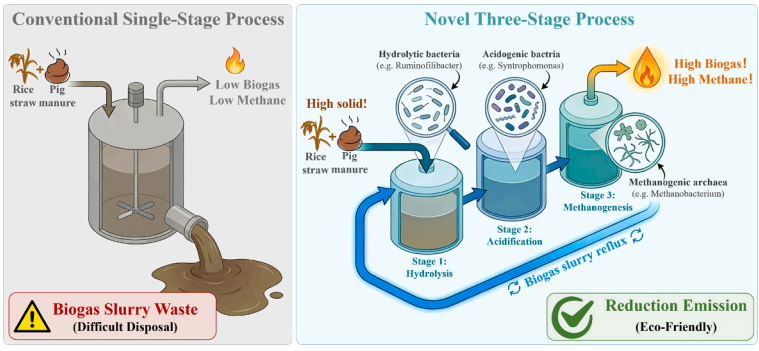
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HIGHLIGHTS

- A three-stage integrated co-digestion (TSAD) process was developed.
- TSAD enriched the stage-specific functional flora by spatial segregation.
- TSAD enabled the reduction of biogas slurry emissions via biogas slurry reflux.
- TSAD enhanced processing capacity and reduced the consumption of clean water.
- HRT = 20 d, TSAD maintained efficient and stable methane production within 120 days.

GRAPHICAL ABSTRACT



ARTICLE INFO

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Agricultural waste
Microbial community
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ABSTRACT

To address the challenges associated with the large-scale production and difficult disposal of biogas slurry in conventional high-solid anaerobic digestion, this study developed a three-stage integrated co-digestion (TSAD) process integrated with biogas slurry reflux (BSR), and assessed its feasibility for stable semi-continuous operation over a 120-day period. Results demonstrate that TSAD enables targeted enrichment among functional microbial communities—achieved through spatially segregated operational stages: aerobic hydrolysis, micro-aerobic acidification, and anaerobic methanogenesis. Under an appropriate hydraulic retention time (HRT) of 20 days, the system achieved sustained high biogas production stability (with a final methane yield of 218 mL/g COD). Crucially, relying on the comprehensive strategy of high-solid feedstock, staged separation and BSR, it enhanced overall treatment efficiency by approximately 2.8-fold relative to conventional single-stage system, while reducing daily water consumption per unit substrate by 83.3 %, thereby substantially mitigating slurry emission. In contrast, reducing the HRT to 10 days boosted short-term biogas production, but also enhanced the mobilization of copper and zinc into the liquid phase and caused rapid accumulation of ammonia nitrogen (up to 4855.1 mg/L) during prolonged operation, ultimately increasing the risk of process instability and potential system failure. These findings provide a theoretical basis for achieving process optimization through staged

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separation and BSR, and underscore the necessity of rational HRT selection and coupling the targeted removal technologies of ammonia nitrogen and heavy metals, so as to ensure sustainable and long-term system performance.

1. Introduction

To achieve the goals of carbon peaking and carbon neutrality, promoting the replacement of fossil fuels with clean energy has become a global consensus (Zhang and You, 2023). Among various strategies, efficient utilization of biomass waste represents a critical pathway (Zhang et al., 2026). Anaerobic digestion (AD), as a mature technology for biomass treatment, enables the conversion of organic matter into clean biogas through microbial synergism. This process offers dual benefits—effective waste management and renewable energy production, and plays a significant role in advancing carbon emission reductions (Awodun et al., 2025; Huang et al., 2019).

As the country with the fastest expansion of AD, China had approximately 100,000 industrial digesters and a total number of about 39.1 million digestion tanks by 2020 (Ni, 2024; Zheng et al., 2020). However, in recent years, it has seen stagnation or even negative growth in project expansion in certain regions (Ni, 2024). Multiple factors contribute to this trend, among which the difficulty in the utilization of biogas slurry (BS) is widely recognized as the most critical bottleneck constraining the sustainable development of biogas projects (Yang et al., 2019; Zeng et al., 2022). As the primary by-product of AD, BS is rich in organic matter and essential nutrients such as nitrogen and phosphorus; however, it may also contain potential contaminants including heavy metals and antibiotic residues (Liu et al., 2021). Although stringent emission standards have been established, compliance requires complex pretreatment procedures and substantial financial investment (Zhang et al., 2020). More critically, even after meeting regulatory requirements, large land areas are still needed for safe application, which many farms lack due to limited surrounding arable land (Sheets et al., 2015). Furthermore, studies indicate that labor costs associated with applying BS and biogas residue are approximately ten times higher than those for chemical fertilizers, significantly reducing farmers' willingness to utilize these materials (Li et al., 2021), which further aggravates the dilemma of BS utilization. These factors collectively contributing to the prevalent phenomenon of "prioritizing construction over operation" in biogas projects and resulting in insufficient resource utilization.

In response to these challenges, reduction and resource-oriented utilization of BS have emerged as key technical directions for the transformation and upgrading of biogas systems. Currently, various approaches are being explored to minimize BS generation and enhance its value. Biogas slurry reflux (BSR) technology facilitates the reuse of organic matter and functional microorganisms by returning a portion of the BS back into the digester (Li et al., 2023). This strategy not only reduces overall BS emission but also enhances methane (CH_4) production and system stability (Li et al., 2023; Yue et al., 2023). Research by Li et al. demonstrates that BSR promotes the establishment of specialized microbial communities, accelerates organic hydrolysis and hydrogen (H_2) production, and delivers both environmental and economic advantages (Li et al., 2020). Nevertheless, conventional single-stage anaerobic digestion (SSAD) processes face inherent limitations: under high organic loading rates (OLR), the rates of hydrolysis/acidification and methanogenesis become mismatched, resulting in a long hydraulic retention time (HRT, typically 20–50 days) for the system (Mahdy et al., 2019; Xu et al., 2023), which is contradictable with the BS reduction requirements.

Owing to the inherent limitations of SSAD systems, two-stage AD configurations and various pretreatment strategies—including aerobic and microaerobic pretreatment, thermal extrusion, and others—have been widely investigated to mitigate process inhibition and enhance biogas yield (Ning et al., 2023; Oluwagbotemi Fasheun et al., 2024; Wang et al., 2024; Wu et al., 2016; You et al., 2024). Nevertheless, these approaches remain susceptible to thermodynamic inefficiencies and potential secondary environmental impacts. Notably, conventional two-stage systems typically integrate hydrolysis and acidogenesis within a single upstream reactor. Given the substantial physiological disparities (e.g., optimal pH) between hydrolytic enzymes and acidogenic bacteria, such spatial co-location engenders kinetic incompatibilities, thereby compromising biochemical conversion efficiency at the initial stage (Tang et al., 2023).

To overcome these constraints, this study introduces a novel three-stage integrated process (TSAD): "aerobic/microaerobic/anaerobic co-digestion combined with BSR." The process aims to decouple AD into three functionally distinct and spatially separated stages, including hydrolysis, acidogenesis, and methanogenesis, thereby eliminating kinetic interference and enabling precise optimization of ecological niches for each functional microbial consortium. Concurrently, combined with BSR, it will achieve the goal of reducing slurry emissions while sustaining high OLR and operational stability. This work will not only offer a new perspective to overcome the long-standing hydrolysis–acidogenesis coupling bottleneck, but also establishes a robust theoretical and technical foundation for advancing next-generation agricultural waste treatment technologies characterized by high conversion efficiency and low emission.

2. Materials and methods

2.1. Inoculum and substrate

Pig manure (PM) used in the experiment was collected from a large-scale swine farm in Pingxiang, Jiangxi, China. After manual screening to remove large soil particles and animal hair, the material was passed through a 60-mesh sieve (particle size < 425 μm) and stored at 4℃ for subsequent use. Rice straw (RS) was sourced from Nanchang, Jiangxi, China. Following mechanical crushing, it was stored through a 60-mesh sieve. The main physicochemical characteristics of RS and PM are presented in Table 1.

The inoculum for the acidification reactor was derived from acetic acid bacterial residue generated during vinegar brewing. A 20-gram sample was sealed in a 100-mesh nylon bag and subjected to pre-adaptation cultivation using effluent from the hydrolysis tank until stable performance was achieved. The inoculum for the methanogenic reactor was obtained from a long-term AD system in Nanchang, where RS served as the primary substrate. Prior to use, it underwent pre-acclimation with effluent from the acidification reactor.

Table 1
Physicochemical properties of RS and PM.

	RS	PM
TS(% ww)	92.3 ± 0.6	34.6 ± 0.6
VS(% ww)	83.9 ± 0.3	22.0 ± 0.5
TC(mg/g)	441.4 ± 3.5	350.5 ± 3.7
TOC(mg/g)	433.2 ± 1.6	331.2 ± 3.2
TN(mg/g)	8.6 ± 0.7	20.5 ± 1.3
Cellulose(% TS)	26.5 ± 0.2	22.2 ± 0.2
Hemicellulose(% TS)	35.1 ± 0.4	20.4 ± 0.2
Lignin(% TS)	7.3 ± 0.1	13.3 ± 0.2

Notes: RS: rice straw; PM: pig manure; TS: Total solid; VS: Volatile solid; ww: Base on the wet weight of the material; TC: Total carbon; TOC: Total organic carbon.

2.2. Equipment description

The schematic designs of the TSAD and SSAD systems are presented in Fig. 1. The TSAD system comprises an aerobic hydrolysis tank (TR1, with a total volume of 14 L and an effective volume of 3 L), followed by a microaerobic acidizing tank (TR2, 1.5 L total volume, 1 L effective volume), and an anaerobic CH_4-producing tank (TR3, total volume of 6 L, effective volume of 5 L). Within TR1, a sieve plate lined with 500-mesh gauze was divided the tank into two functional zones: an upper material storage zone with top-space pressure equalization (approximately 9 L), and a lower hydrolysis liquid collection zone (with a total volume of 5 L and an effective volume of 3 L). The SSAD system consists of a continuous stirred-tank reactor (CSTR) (SR, total volume was 5 L, effective volume was 4 L). All reactors are equipped with external circulating water jacket layers for temperature control, maintained at 37℃ using a thermostatic water bath (SHZ-B, China Boxun Medical Biological Instrument Co., LTD). Gas outlets from SR, TR2 and TR3 are connected to gas collection bags, and these reactors are fitted with intermittent stirring devices operating at 20 rpm, programmed to stir for 5 min at 8-hour intervals. The material transfer between reactors is achieved through peristaltic pumps.

2.3. Experimental design

2.3.1. Equipment operation and feed composition of each reactor

The operation of the TSAD system follows this process: Firstly,

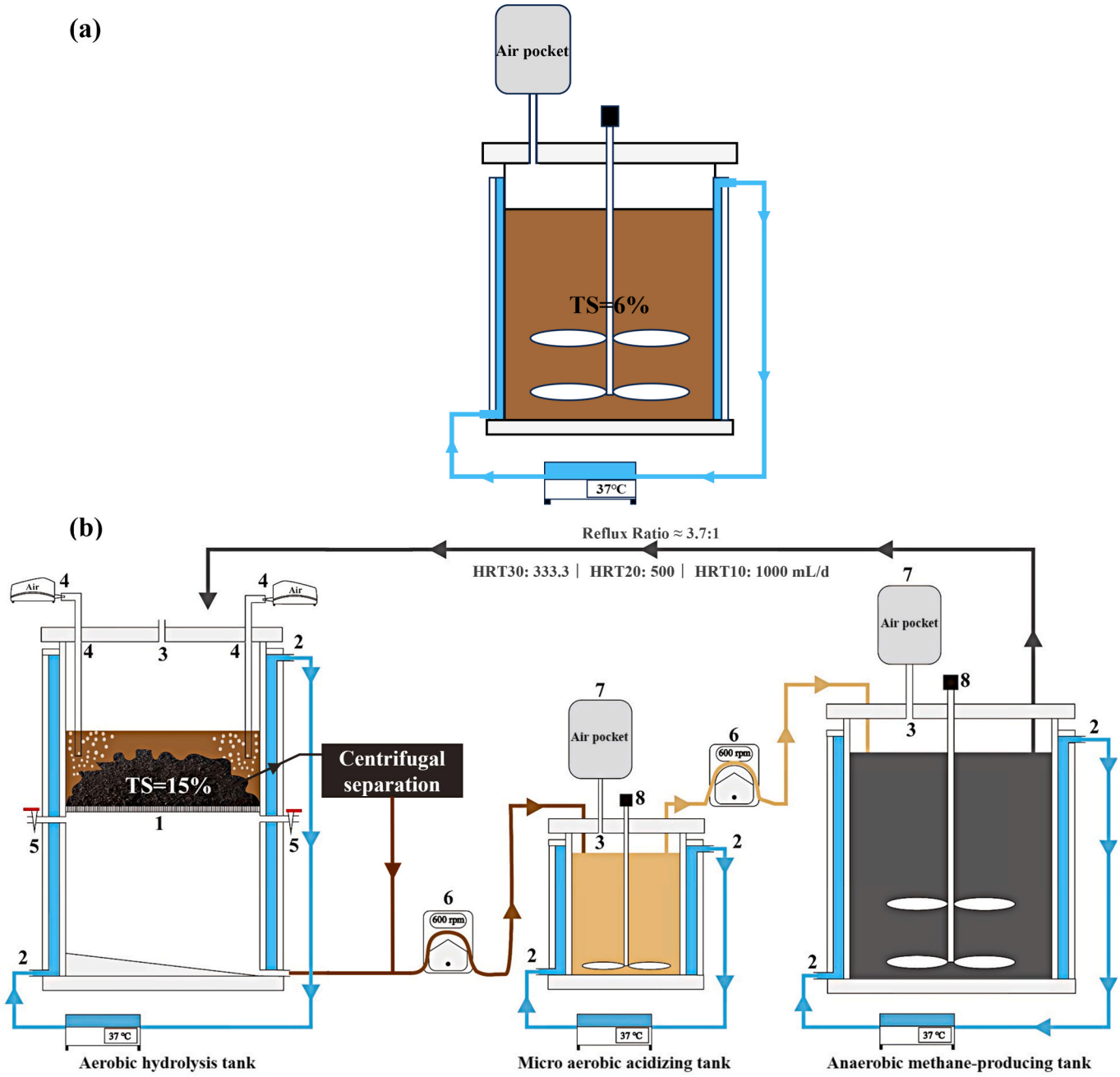


Fig. 1. Equipment design of SSAD (a), and TSAD (b) systems. Notes: The blue arrow indicates the 37℃ constant temperature water circulation path. The brown arrow indicates the direction of BS transportation from TR1 to TR2. The yellow arrow indicates the direction of BS transportation from TR2 to TR3. The black arrow indicates the direction of BS reflux from TR3 to TR1. In the design drawing (b), 1 represents a perforated sieve plate; 2 is the inlet and outlet of the water bath jacket. 3 is the air outlet of the device. 4 is the aeration device; 5 is the balanced air pressure outlet; 6 is a peristaltic pump; 7 is the gas collection bag; 8 is the stirring device.

thoroughly mixed feedstock was put into the upper section of the sieve plate of TR1. By closing the lower outlet valve, the materials were temporarily retained in the upper chamber. Subsequently, an external air pump was activated, introducing air through the aeration port, while exhaust gases were expelled via the exhaust outlet—thereby enabling daily aerobic hydrolysis. Aeration was conducted intermittently—5 min of working followed by 5 min of stopping—with a volumetric airflow rate of 2000 mL-air/min during operation. When liquid–solid separation was required, the pressure relief valve was opened to equalize the internal pressure. Under gravitational force, the hydrolyzed liquid flowed along the 15° inclined base of the tank toward the outlet and was then transferred to TR2 using a peristaltic pump. Upon completion of acidification, the treated material was pumped into TR3 for further processing. Finally, the effluent from TR3 was combined with fresh feedstock and recirculated back to TR1, thereby achieving the recycling of liquid slurry. The SSAD system continuously fed in and out in accordance with the conventional CSTR mode.

To ensure stable system operation, the daily feed compositions for each reactor in the TSAD system are specified as follows: The daily feedstock for TR1 consisted of a mixture of PM and RS at a mass ratio of 1:1, with liquid slurry recirculated from TR3, and supplementary distilled water. More details on the reflux liquid slurry are provided in the supplementary materials (see Supplementary Materials). The daily feed to TR2 comprised two components: (i) the hydrolysate obtained by centrifugation (5,000 rpm) of the equal amount of mixture discharged from the upper part of TR1 (with a liquid separation rate of approximately 40 %); and (ii) the residual portion of the hydrolysate obtained from the lower-layer slurry of TR1 following depressurization. The daily feed of TR3 was the complete effluent from TR2. The SSAD system employed an identical ratio of PM to RS as TR1 but substituted distilled water for BS and maintained the TS concentration at 6 % to mitigate potential ammonia inhibition (Wijesinghe et al., 2021).

2.3.2. Comparative experiments of SSAD and TSAD system under different HRTs

Throughout the entire experimental period, the total HRT was kept consistent both the TSAD and SSAD systems. Based on prior research findings (Xiang et al., 2022), the experiment was divided into multiple HRT stages. The experiments involving varying HRTs were conducted sequentially: for each HRT, the system was operated continuously until steady-state conditions (at least 2 to 3 HRTs), after which the system was transitioned to the next designated HRT. The specific feed parameters of each reactor in the SSAD and TSAD systems at different HRTs are shown in Table 2.

2.3.3. Long-term operational stability test

Following the comparative HRT experiment, a 120-day continuous operation test was conducted to assess the long-term operational stability of the two systems. Specifically, the SSAD system was operated at an HRT of 30 days, whereas the TSAD system was evaluated under two distinct HRT conditions—20 days and 10 days.

Table 2
Setting of operating parameters for SSAD and TSAD processes.

Operational phase	Process	Reactor	HRT (day)	Daily feed volume	
				Feed mixture	Feed liquid
Phase 1 (0–60 days)	SSAD	SR	30	133.3 g	–
	TSAD	TR1	12	250 g	–
		TR2	3	–	333.3 mL
		TR3	15	–	333.3 mL
Phase 2 (60–100 days)	SSAD	SR	20	200 g	–
	TSAD	TR1	8	375 g	–
		TR2	2	–	500 mL
		TR3	10	–	500 mL
Phase 3 (100–130 days)	SSAD	SR	10	400 g	–
	TSAD	TR1	4	750 g	–
		TR2	1	–	1000 mL
		TR3	5	–	1000 mL

2.4. Sampling and chemical analysis

During the HRT comparison experiment, digested samples were collected daily from the sampling port of each reactor. For the long-term stability experiment, samples were obtained at three critical time points: system startup (Day 0), mid-operation (Day 60), and end-of-operation (Day 120). Measurement of TS, VS, TC, TOC, volatile fatty acids (VFAs), soluble chemical oxygen demand (sCOD), total ammonia nitrogen (TAN), and lignocellulosic components (including lignin, cellulose, and hemicellulose) was analyzed referring to the previous report (Xiang et al., 2022).

Generated biogas was collected using gas collection bags, and the total volume was recorded using a wet gas flowmeter (LMF-1, Jinzhiye Instrument & Equipment Co., Ltd., Beijing, China). The concentrations of CH_4 and carbon dioxide (CO_2) in the biogas were analyzed using a portable biogas analyzer (Biogas 5000, Geotech, UK).

The concentrations of heavy metals, including Copper (Cu) and Zinc (Zn), in both liquid and solid fraction of the effluent were determined by inductively coupled plasma optical emission spectrometer (ICP-OES, OPTIMA 8000, PerkinElmer, USA).

2.5. Microbial structure analysis

The microbial diversity of fungi and bacteria in TR1, bacteria and archaea in TR2 and archaea in TR3 were measured. Sequencing was conducted by Shanghai Majorbio Technology Co., Ltd. The specific methods are referred to in previous report (Xiang et al., 2022). Primers for amplifying the gene were as follows: ITS1F (5′-CTTGGTCATTTAGAGGAAGTAA-3′) and ITS2R (5′-GCTGCGTTCTTCATCGATGC-3′) were used to amplify the ITS1 region of the fungal ITF gene. The V3-V4 region of bacterial 16S rRNA was amplified using 338F (5′-ACTCCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′). The V4-V5 regions of archaea were amplified using 524F10extF (5′-TGYCAGCCGCCGCGGTAA-3′) and Arch958RmodR (5′-YCCGGCGTTGAVTCCAATT-3′).

2.6. Carbon flow analysis

Carbon flow analysis was calculated based on the total mass, TS, TC and TOC change recorded during the experiment. The detailed calculation method is shown in the supplementary materials (see Supplementary Materials).

2.7. Statistics

The obtained data were organized using Microsoft Excel 2016. Graphs were plotted using Origin 2024. The data are expressed as means ± standard deviations (M ± SD), based on at least three technical replicates per sample. The sequencing results were analyzed in the Majorbio Cloud Platform (https://www.majorbio.com). The Circos map of microbial community structure was drawn through the online platform (https://mkweb.bcgsc.ca/tableviewer/visualize/) based on the method of (Krzywinski et al., 2009). No hypothesis testing (e.g., ANOVA) was performed due to the absence of biological replicates (i.e., single reactor sequential operation).

3. Results and discussion

3.1. Operational performance of TSAD and SSAD processes

3.1.1. Aerobic hydrolysis stage of TSAD versus SSAD

Regarding solid degradation, when the HRT of the aerobic hydrolysis unit in the TSAD system was maintained at 12 days, the degradation

rates of TS and VS reached 27.9 % and 36.6 %, respectively (Fig. 2a). Under these conditions, the VS/TS ratio was 75.2 %, indicating relatively high microbial activity, which facilitates the effective breakdown of volatile solids and contributes to system stability. In contrast, in the SSAD system, reducing the HRT from 30 days to 10 days resulted in insufficient utilization of readily degradable organic matter, as evidenced by declining TS and VS degradation rates and an increase in the VS/TS ratio from 78.5 % to 83.8 % (Fig. 3a). With respect to carbon conversion, a reduction in HRT from 30 to 10 days in the SSAD system led to a decrease in solid-phase inorganic carbon (IC) content from 40 mg/g to 12 mg/g, suggesting that a substantial portion of TOC remained unconverted in the solid fraction. Conversely, in the TSAD system with an HRT of 4 days in the aerobic hydrolysis stage, although TC and TOC levels in the solid phase increased due to reduced carbon transfer, the TOC proportion in the liquid slurry reached 94.7 %. This indicates that nearly all organic carbon was successfully transferred into the liquid phase in bioavailable forms, thereby providing abundant substrate for the subsequent anaerobic digestion stage. This outcome highlights a marked improvement over the limited carbon utilization observed in the SSAD system.

Furthermore, when the HRT of the SSAD system was reduced to 10 days, significant accumulation of dissolved organic matter occurred. The effluent concentrations of sCOD and VFAs reached 5,672 mg/L and 2,974 mg/L, respectively (Fig. 3b), with VFA levels substantially exceeding the recommended threshold for stable operation (500 ~ 2000 mg/L) (Xu et al., 2023). This observation aligns with the findings of

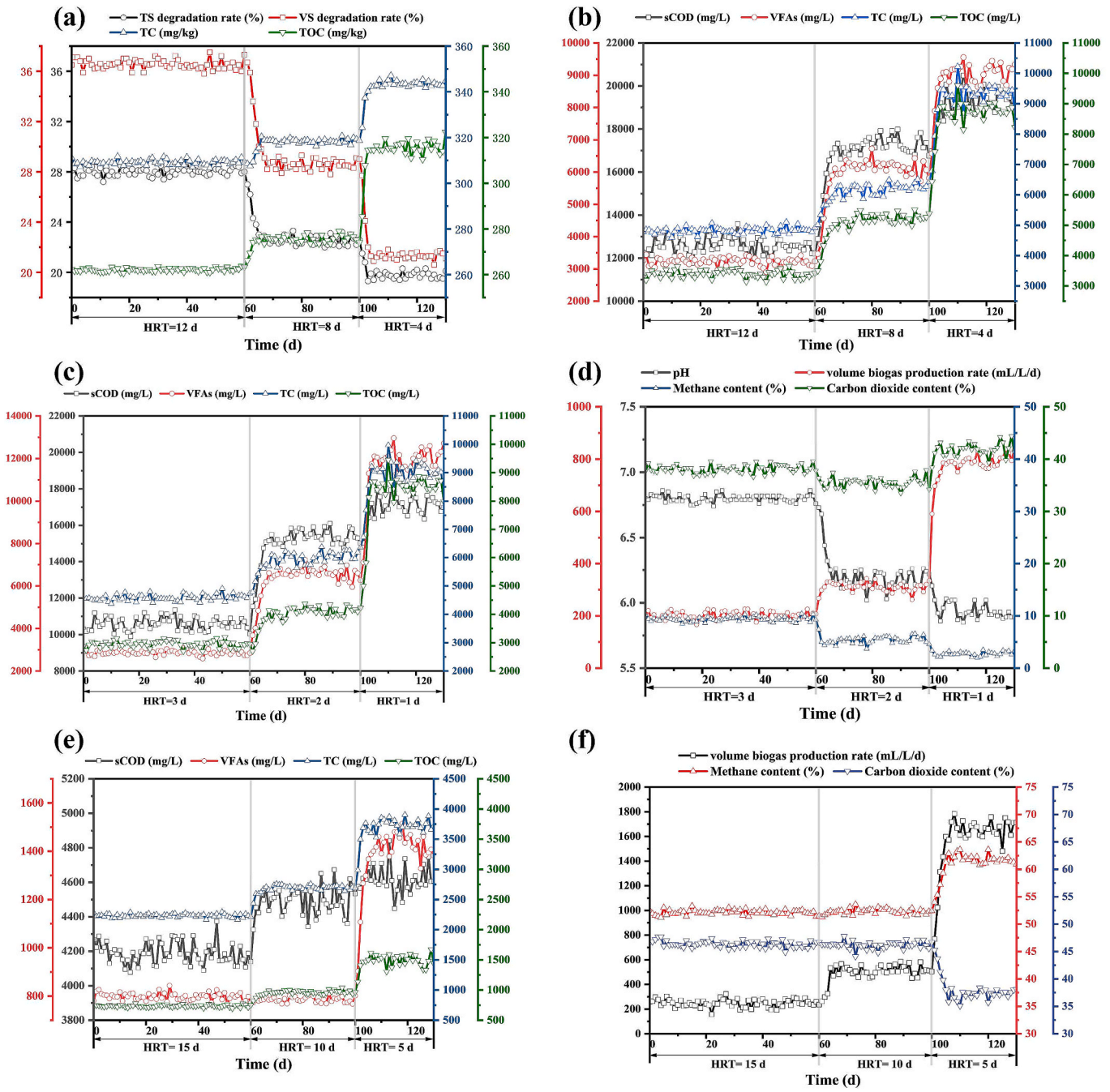


Fig. 2. Changes in TS and VS degradation rates, TC and TOC of solid residue (a), and sCOD, VFAs, TC, TOC of liquid slurry (b) in TR1; sCOD, VFAs, TC, TOC of liquid slurry (c), and pH, volume biogas production rate, CH_4 and CO_2 content (d) in TR2; sCOD, VFAs, TC, TOC of liquid slurry (e), and volume biogas production rate, CH_4 and CO_2 content (f) in TR3.

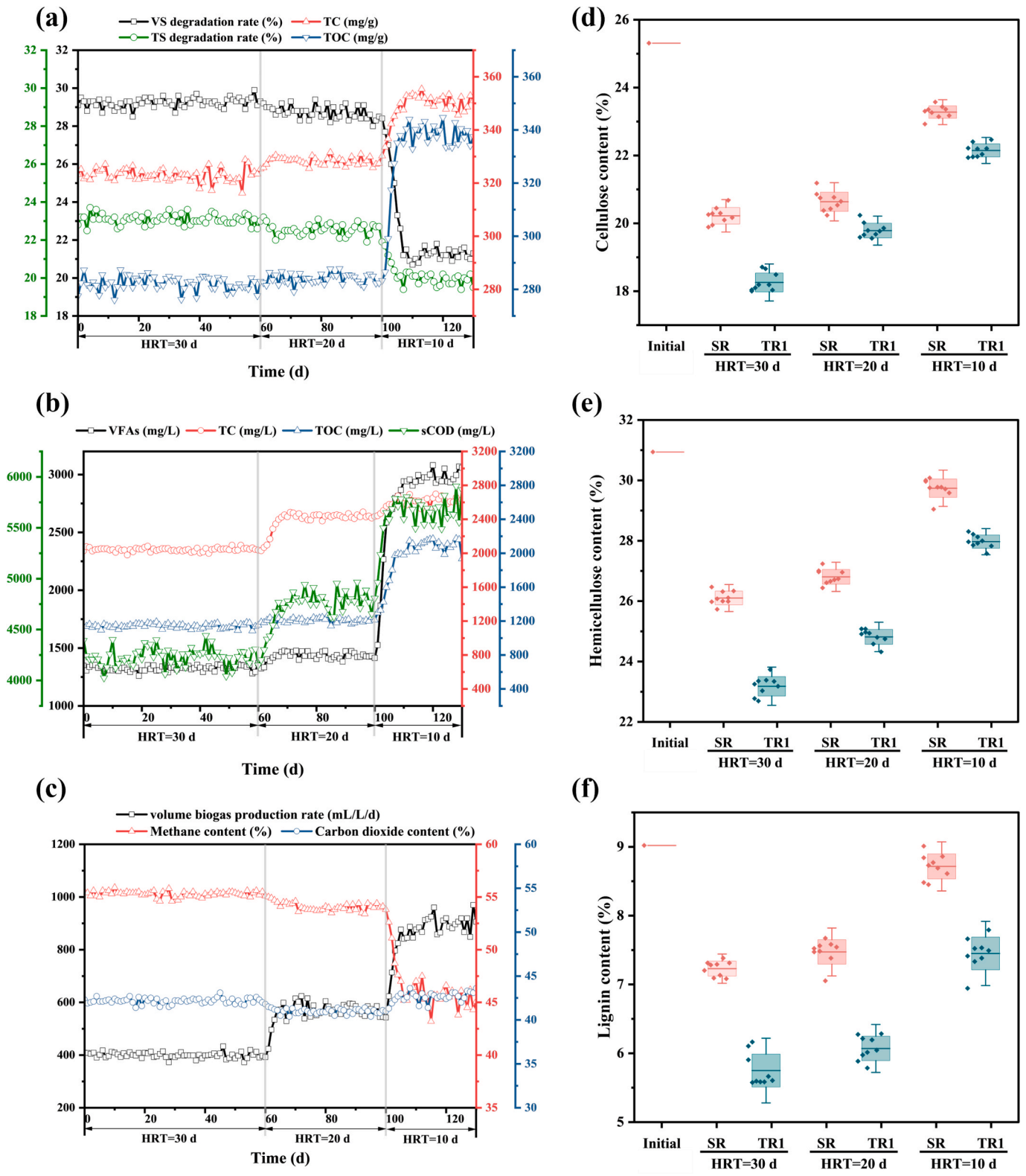


Fig. 3. Changes in TS and VS degradation rates, TC and TOC of solid residue (a), sCOD, VFAs, TC and TOC of liquid slurry (b), and volume biogas production rate, CH_4 and CO_2 content (c) in the SSAD system; Comparison of cellulose (d), hemicellulose (e), and lignin (f) contents in the liquid slurry of SR and TR1.

Trujillo-Reyes et al. (2023) and Wu et al. (2022), who reported that increasing OLR easily lead VFA accumulation and operational instability were frequently observed in the SSAD system. In comparison, during the aerobic hydrolysis stage of the TSAD process, even when the HRT was shortened from 12 to 4 days—resulting in 84 % and 215 % increases in sCOD and VFAs, respectively (Fig. 2b)—the accumulation remained within controllable limits and was functionally beneficial. The elevated concentration of dissolved organics, particularly the almost completely organic carbon (with TOC accounting for 94.7 %), precisely demonstrates that aerobic hydrolysis effectively transforms solid substrates

into high-quality, microbially accessible feedstock for the downstream anaerobic stage, thus preventing the detrimental accumulation typically caused by inadequate conversion capacity in the SSAD system. These findings are consistent with those reported by (Rennuit et al., 2018), confirming that aerobic pretreatment effectively improves the bioaccessibility and biodegradation efficiency of complex organic substrates.

The lignocellulose degradation performance in the SSAD system and aerobic hydrolysis stages of TSAD system are illustrated Fig. 3d-3f. The results show that, under varying HRT conditions, TR1 (representing the TSAD system) achieves higher degradation rates of cellulose, hemicellulose, and lignin compared to SR (SSAD), with only 40 % of the treatment time required. Notably, at an HRT of 10 days, TR1 exhibited degradation rate enhancements of 35.2 %, 46.4 %, and 71.0 % for cellulose, hemicellulose, and lignin, respectively, relative to SR. These results underscore the superior short-term efficiency of aerobic hydrolysis in breaking down recalcitrant lignocellulosic components, thereby ensuring a sufficient supply of bioavailable substrates for the subsequent CH_4 production phase. This finding aligns with the results reported by Zhang et al., 2019a, who demonstrated that aerobic hydrolysis markedly disrupted the lignin structure across all anatomical components of corn stalks, thereby increasing the specific CH_4 yield (per unit volatile solids) by 8–35.02 % relative to the untreated control and substantially reducing the overall fermentation duration.

3.1.2. Microaerobic acidification stage of TSAD versus SSAD

With respect to liquid-phase metabolites, the microaerobic acidification reactor has demonstrated an active regulatory capability for dissolved organic matter. When the HRT of the acidification stage is maintained at 3 days, the effluent VFA concentration (2879 mg/L) is lower than that of the influent (3239 mg/L), indicating net VFA consumption (Fig. 2c). However, when the HRT is reduced to 1 day, the reactor actively enriches VFAs to a concentration of 11,987 mg/L—1.32 times higher than the influent—demonstrating its adaptive metabolic control. This flexible regulatory capability ensures the provision of substrates with appropriate concentration and optimized composition for the subsequent methanogenic stage, which is fundamentally different from the malignant accumulation of VFAs in the SSAD system.

The distinction between the two systems becomes even more pronounced in terms of operational conditions and product profiles. While the SSAD system was accumulating VFAs, it was accompanied by the decrease of CH_4 content to 45.7 % and system fluctuations, which reflected the inherent contradiction that hydrolysis, acidification and CH_4 production inhibited each other in the same space (Yang et al., 2023). In contrast, in the microaerobic acidification stage of the TSAD system, shortening the HRT from 3 to 1 day leads to enhanced VFAs concentration, a corresponding decrease in system pH from 6.76 to 5.89, and an notable increase in volume biogas production rate (Fig. 2d), which may be attributed to the generation of CO_2 and H_2 during acidogenesis and acetogenesis (Al-Rubaye et al., 2020; Gujer and Zehnder, 1983; Parra-Orobio et al., 2020). Moreover, the CH_4 production observed during this stage is likely due to the introduction of a limited population of methanogenic archaea via BSR (as detailed in Section 3.2), while the low CH_4 content may be associated with inhibitory effects caused by acid accumulation under low HRT conditions (Dzofou Ngoumelah et al., 2023; Guo et al., 2025; Nguyen and Khanal, 2018).

3.1.3. Anaerobic methanogenic stage of TSAD versus SSAD

The anaerobic methanogenic stage, serving as the final phase of the TSAD system, exhibits high substrate conversion efficiency and superior biogas quality. When the HRT was reduced from 10 to 5 days, the volume biogas production rate increased from 522 mL/L/d to 1665 mL/L/d. The combined content of CH_4 and CO_2 in the biogas consistently exceeded 97.5 % (Fig. 2f), thereby reducing the difficulty of subsequent biogas purification. Notably, as the HRT decreased from 15 to 5 days, the CH_4 content rose from 52.5 % to 61.9 %, whereas the CO_2 content declined from 45.5 % to 37.2 %. This phenomenon may be attributed to the change in the archaeal community structure during this stage (see Section 3.2 for details). This shift may be attributed to enhanced H_2 production driven by elevated organic acid concentrations in the feed, which promotes the hydrogenotrophic methanogenesis pathway (Zhu et al., 2020, Zhu et al., 2019), resulting in increased CH_4 formation through CO_2 consumption. In contrast, the SSAD system achieved a volume biogas production rate of 401.2 mL/L/d at an HRT of 30 days, with a CH_4 content of approximately 55.2 %. Although the volume biogas production rate increased to 904.9 mL/L/d when the HRT was reduced to 10 days, the CH_4 content dropped markedly to 45.7 %, indicating challenges in maintaining stable biogas composition under higher OLR. Furthermore, the anaerobic methanogenic stage of TSAD system achieved removal efficiencies of 73.1 %, 88.1 %, 58.7 %, and 82.6 % for sCOD, VFAs, TC, and TOC, respectively (Fig. 2e). Despite the significantly shortened HRT, the TSAD system did not impose a substantial burden on subsequent treatment processes, demonstrating superior operational flexibility and conversion efficiency compared to the SSAD configuration.

3.2. Functional response of microbial community structure to TSAD system

The relative abundances of different bacterial communities of different stages at the genus level were shown in Fig. 4. During the hydrolysis stage, the fungal community was predominantly composed of *Candida*, *Geotrichum*, and *Trichosporon*. Notably, the relative abundance of *Candida* increased by 17-fold (from 2.34 % to 42.3 %) under short HRT conditions (Fig. 4a). Previous studies have demonstrated that *Candida* plays a pivotal role in the rapid degradation of lignocellulosic materials (Li et al., 2017a), which aligns with the observed high lignocellulose degradation efficiency during this stage (Fig. 3d-3f). Within the bacterial community, hydrolytic and acidogenic genera such as *Ruminofilibacter*, *Prevotella*, *Ruminiclostridium*, and Prevotellaceae_UCG-001 were dominant (Fig. 4b). *Prevotella* and Prevotellaceae_UCG-001 belong to the family Prevotellaceae, which has been reported to facilitate the hydrolysis and acidification of straw substrates (Lazuka et al., 2015; Jose et al., 2017). *Ruminofilibacter* and *Ruminiclostridium* are recognized as key functional microorganisms involved in the conversion of carbohydrates into small-molecule organic acids (Azman et al., 2015; Kröber et al., 2009), indicating their synergistic contribution to the breakdown of insoluble organic matter and the generation of soluble intermediates during hydrolysis.

In the acidification stage, the bacterial community structure exhibited remarkable shifts in response to changes in HRT (Fig. 4c). At a total HRT of 10 days, *Acetobacter* (11.0 %), *Syntrophomonas* (12.7 %), *Ruminiclostridium* (6.5 %), *Mobilitalea* (5.6 %), and *Butyrivibrio* (9.2 %) emerged as dominant genera. These taxa are functionally associated with the production of short-chain fatty acids (e.g., acetic and butyric acids) and participate in H_2 and acetate formation pathways, collectively supporting the stable accumulation of high concentrations of VFAs (An et al., 2018; Krusong et al., 2022; Nielsen et al., 2018). Additionally, shortening the HRT led to a 3.5-fold increase in the relative abundance of *Methanobacterium* (from 12.3 % to 54.7 %) and a 1.7-fold increase in that of *Methanocorpusculum* (from 6.70 % to 11.4 %), both of which are hydrogenotrophic methanogens; conversely, the relative abundance of the acetoclastic methanogen *Methanosarcina* declined from 70.2 % to 28.5 % (Fig. 4d). Rośen C, J.U., 2005 have demonstrated that acetoclastic methanogens exhibits heightened sensitivity to low pH and elevated VFA concentrations, whereas certain hydrogenotrophic methanogens display comparatively greater acid tolerance. Furthermore, Tang et al. (2023) reported that in continuous-culture systems, when the retention time of microbial cells is lower than their limit value, inducing a dynamic stress condition that compromises metabolic activity—potentially resulting in markedly reduced or even arrested CH_4 production CH_4 production. This might explain the

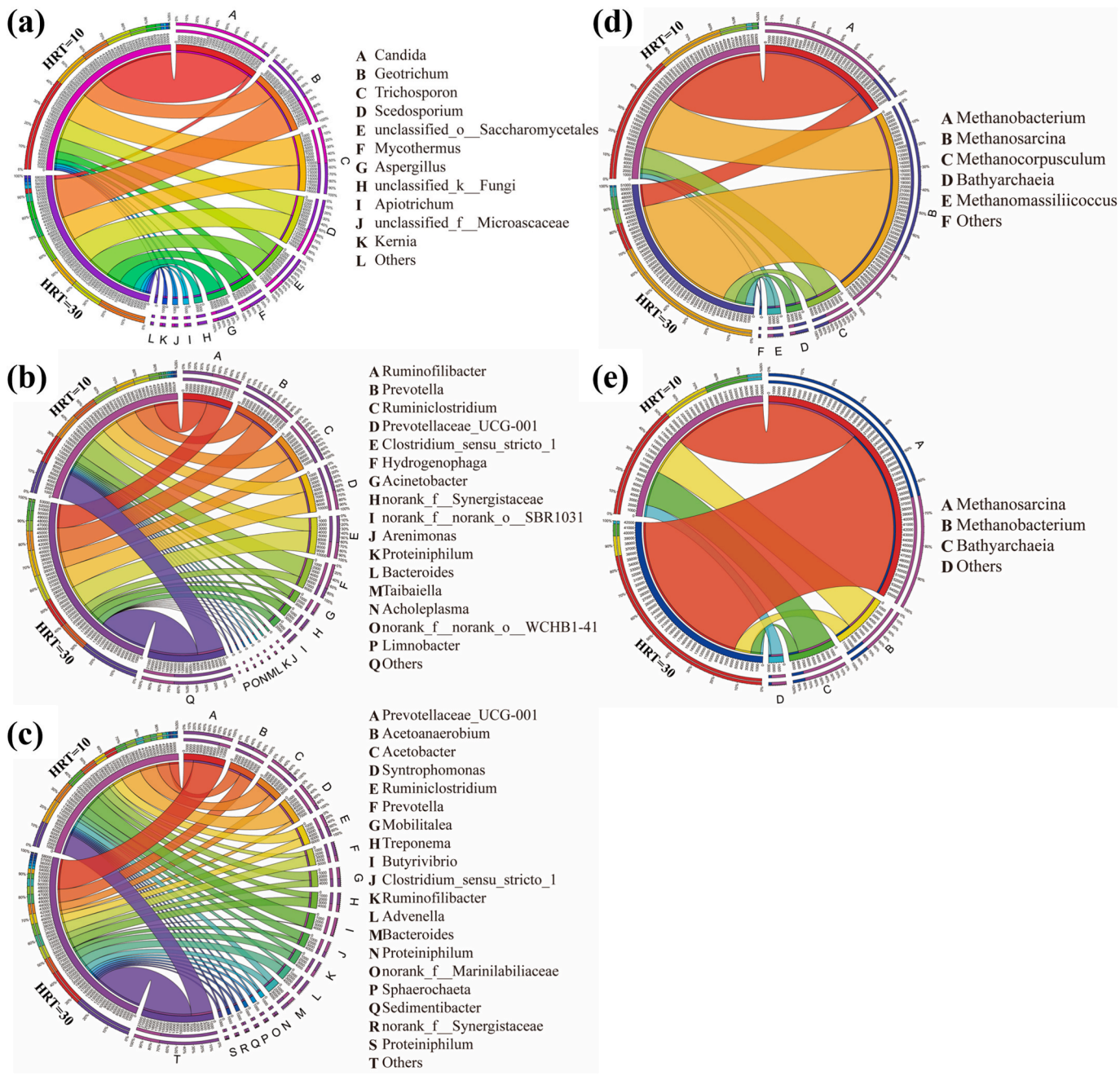


Fig. 4. Microbial diversity of fungi in the hydrolysis stage (a); Microbial diversity of bacteria in the hydrolysis stage (b) and the acidification stage (c); Microbial diversity of methanogenic archaea in the acidification stage (d) and the methanogenic stage (f).

extremely low CH_4 production observed during this stage.

The archaeal community structure in the methanogenic stage showed a pronounced response to HRT variations (Fig. 4e). When HRT was reduced from 30 to 10 days, the relative abundance of acetoclastic *Methanosarcina* declined from 85.5 % to 52.3 %, whereas the abundances of hydrogenotrophic *Methanobacterium* and *Bathyarchaeota*—key contributors to the hydrogenotrophic methanogenesis pathway—increased by 1.6-fold (from 8.22 % to 21.3 %) and 2.9-fold (from 5.04 % to 19.7 %), respectively (He et al., 2016; Schöne et al., 2022; Zhang et al., 2019b). This structural shift is likely attributable to inherent differences in the kinetic and thermodynamic properties among functional microbial guilds. Studies have indicated that hydrogenotrophic methanogens and acidogenic bacteria share comparable kinetic traits, including higher half-saturation concentration (K_s) and maximum microbial growth rate (μ_{max}), showing rapid proliferation and robust tolerance to elevated organic loading (Pavlostathis and Giraldo-Gomez 1991; Tang et al., 2023). Consequently, *Methanobacterium* remained effectively retained and active under shortened HRT conditions. In contrast, acetoclastic methanogens—characterized by longer generation times—are more susceptible to washout when HRT was reduced, resulting in a decrease in their relative abundance.

In summary, the community structure in this process differs significantly from that in conventional SSAD systems, where various functional microorganisms—such as hydrolytic and acid-producing bacteria from Firmicutes and Bacteroidetes, along with archaea like Methanosaeta—coexist and are prone to metabolic imbalance (Wang et al., 2017; Wu et al., 2016; Xiao et al., 2022; You et al., 2024). This system establishes an optimized environment for distinct microbial populations through physical compartmentalization. Specifically, the hydrolysis stage enriches fungi (e.g., *Candida*) and hydrolytic bacteria

(e.g., *Ruminofilibacter*), which initiate the disruption of the lignocellulosic structure. Subsequently, the acidification stage promotes the growth of acidogenic bacteria such as *Syntrophomonas* and *Acetobacter*, facilitating efficient conversion of the substrate into VFAs. Consequently, the methanogenesis stage receives stable and high-quality substrate, thereby supporting an optimized archaeal consortium—initially dominated by acetoclastic *Methanosarcina*, but progressively transitioning toward a balanced and cooperative community structure of acetoclastic, hydrogenotrophic, and auxiliary methanogens—which ultimately enables biogas production with elevated CH_4 content (61.9 %).

3.3. Analysis of carbon flow

This study conducted a comparative analysis of carbon flow in SSAD and TSAD systems (Fig. 5), to evaluate resource utilization and reduction efficiency across different processes. In the SSAD process, as HRT decreased from 30 to 10 days, the proportion of gaseous carbon (CH_4 and CO_2) converted declined, while the fraction of liquid carbon (in BS) increased. This indicates reduced biogas production efficiency and a higher burden on downstream BS treatment. Such phenomenon is commonly observed in conventional SSAD system, particularly under shortened HRTs, the slow-growing methanogens are prone to washout, leading to incomplete VFA consumption and consequent carbon accumulation in the liquid phase (Mao et al., 2015). In contrast, in the TSAD process, as the total HRT was shortened from 30 days to 10 days, the carbon distribution in the hydrolysis stage changed: the proportion of carbon lost as fugitive gas decreased from 34.6 % to 17.4 %, whereas the carbon content in the hydrolysate increased from 10.0 % to 18.7 %. This shift suggests enhanced transfer of carbon from volatile forms into the liquid phase, thereby providing more available substrates for subsequent stages. Under a 10-day HRT, the total carbon entering the methanogenesis stage increased by 0.9 times compared to the 30-day condition. Ultimately, 6.7 % of the initial carbon was converted into CH_4, resulting in a CH_4 production 2.6 times higher than that under prolonged HRT. Although the CH_4 conversion efficiency of the TSAD system at a 10-day HRT was 63 % lower than that of the SSAD system at 30 days, its CH_4 production rate per unit time was substantially improved, with comparable residual carbon levels in solid and liquid phases. The observed difference primarily arises from aerobic carbon loss during the hydrolysis stage—specifically, partial oxidation of organic carbon to CO_2 via aerobic microbial respiration. Nevertheless, extant literature indicates that aerobic pretreatment or microaerobic hydrolysis, despite modest organic carbon losses, significantly accelerates both hydrolytic enzyme synthesis and the depolymerization of recalcitrant biopolymers (Huiliñir et al., 2023). Thus, this carbon "input" has resulted in an enhancement of the overall processing throughput and system stability, highlighting the process's comprehensive performance advantages under high loading conditions.

3.4. Potential and limitations of TSAD process

The concentration of TAN and heavy metal (Cu and Zn), biogas production potential during the long-term operation of the SSAD and TSAD systems are presented in Table 3.

The accumulation of ammonia nitrogen constitutes a critical limiting factor for the long-term stable operation of AD systems (Chen et al., 2015; Huang et al., 2018). In the SSAD system operated at an HRT of 30 days, the TAN concentration increased from 494.4 mg/L on Day 0 to 2694.4 mg/L on Day 120, while CH_4 yield rose steadily from 79.7 mL/g COD to 95.7 mL/g COD—indicating robust process stability. By contrast, in the TSAD system operated at an HRT of 20 days over 120 days, TAN accumulated to 3290.4 mg/L; yet the system sustained a high volumetric biogas production rate of 589 mL/L/d, with CH_4 content remaining stable at 55.6 %. Concurrently, the specific CH_4 yield (based on COD) increased from 179.8 mL/g COD on Day 0 to 236.4 mL/g COD on Day 60, and remained elevated at 218.0 mL/g COD on Day 120—further confirming operational resilience under moderate ammonia stress. This might be attributed to progressive microbial adaptation to elevated ammonia concentrations, indicating that the TSAD system exhibits certain ammonia tolerance within this operational load range. As documented in prior literature, TAN concentrations exceeding 3000 mg/L markedly enhance ammonium ion toxicity independent of pH, thereby compromising process stability (Wu et al., 2016). In this study, when the HRT was further reduced to 10 days, TAN in the TSAD system rose to 3081.2 mg/L after 60 days and escalated to 4855.1 mg/L after 120 days. Concurrently, the biogas production rate declined sharply—from 1423 mL/L/d to 535 mL/L/d—and CH_4 content fell below 40 %, and the CH_4 yield dropped precipitously from 248.3 mL/g COD on Day 60 to 60.2 mL/g COD on Day 120, signifying severe

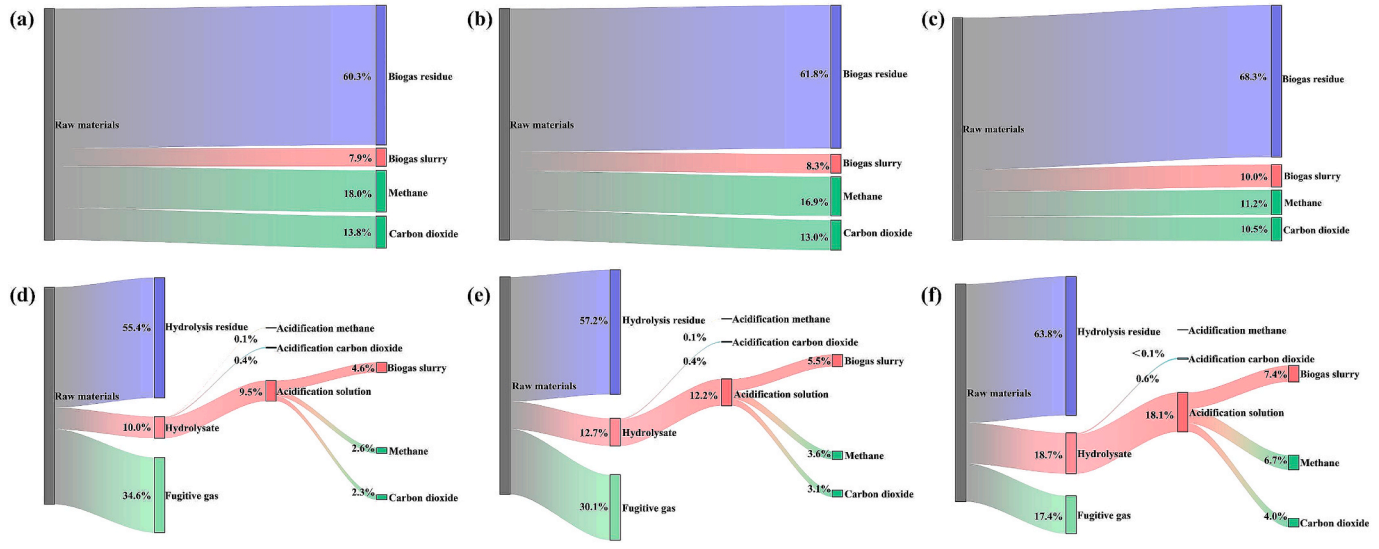


Fig. 5. Carbon flow analysis of SSAD processes at HRT of 30 days (a), 20 days (b), and 10 days (c). Carbon flow analysis of the TSAD process with total HRT of 30 days (d), 20 days (e), and 10 days (f). Notes: In the TSAD process, the effluent from hydrolysis is a solid–liquid mixture. After separation, only a portion of the liquid enters the acidification stage; the carbon in the unrecovered liquid phase is included in the hydrolysis residue. "Acidification methane" and "Acidification carbon dioxide" refer to gases produced during the acidification stage.

Table 3
Operational performance of the SSAD and TSAD processes during the long-term stability test.

	SSAD HRT = 30 d			TSAD HRT = 20 d			HRT = 10 d		
	Day 0	Day 60	Day 120	Day 0	Day 60	Day 120	Day 0	Day 60	Day 120
TAN (mg/L)	494.4 ± 5.2	1060.8 ± 4.1	2694.4 ± 3.4	565.6 ± 6.0	1854.4 ± 2.3	3290.4 ± 8.8	579.2 ± 5.7	3081.2 ± 2.3	4855.1 ± 4.5
Cu in solid residues (mg/kg)	38.6 ± 0.5	39.7 ± 0.5	37.9 ± 0.6	87.2 ± 0.9	89.2 ± 0.9	88.2 ± 1.5	80.3 ± 2.8	79.2 ± 0.6	81.1 ± 3.2
Cu in liquid slurry (mg/L)	4.0 ± 0.4	3.7 ± 0.4	4.1 ± 0.7	9.2 ± 0.5	10.5 ± 0.5	10.8 ± 0.5	18.6 ± 0.8	20.5 ± 0.6	21.2 ± 1.0
Zn in solid residues (mg/kg)	91.0 ± 2.6	89.6 ± 1.4	88.8 ± 2.5	117.1 ± 2.2	122.5 ± 2.3	122.5 ± 1.5	109.6 ± 2.0	114.2 ± 2.1	107.4 ± 1.0
Zn in liquid slurry (mg/L)	8.0 ± 0.3	7.8 ± 0.3	7.8 ± 0.5	12.8 ± 0.4	14.2 ± 0.2	15.3 ± 0.5	28.2 ± 1.1	30.5 ± 1.0	29.1 ± 1.7
Volume biogas production rate (mL/L/d)	385	412	426	542	623	589	896	1423	535
CH_4 content (%)	50.1	55.3	54.4	51.5	57.9	55.6	59.4	60.1	38.7
CH_4 yield (mL/g COD)	79.7	94.1	95.7	179.8	236.4	218.0	156.0	248.3	60.2
COD conversion (%)	20.9	26.9	27.5	51.4	67.6	62.3	44.6	71.0	17.2

Notes: The Cu/Zn concentration in solid residues is calculated based on dry matter, while the Cu/Zn concentration in liquid slurry is calculated based on volume. In the TSAD system, the volume biogas production rate, CH_4 yield, and COD conversion are all calculated based on the effective volume and the COD concentration of the feedstock of TR3.

inhibition of methanogenesis. Collectively, these findings demonstrate that under ultra-high organic loading conditions—induced by excessively short HRT—the TSAD system is highly susceptible to functional failure due to rapid ammonia accumulation. Therefore, from the standpoint of long-term operational robustness, an HRT of 20 days is recommended for the TSAD system, as it balances high treatment efficiency with effective mitigation of ammonia-induced instability.

The distribution of heavy metals in the system effluent warrants careful consideration. In this study, Cu and Zn were predominantly retained in the solid residue of both the SSAD and TSAD systems, consistent with prior findings attributing heavy metal immobilization to complexation and adsorption onto organic matter (Carliell-Marquet and Wheatley, 2002). Notably, the concentrations of Cu and Zn in both the solid residue and liquid slurry of the TSAD system exceeded those observed in the SSAD system. This trend aligns with Ni et al.'s findings on enhanced heavy metal accumulation during anaerobic digestion when BS is recirculated (Ni et al., 2017). Moreover, shortening the HRT from 20 to 10 days led to increased Cu and Zn concentrations in the liquid slurry of the TSAD system. This increase is likely attributable to elevated hydraulic shear stress under higher OLRs. In anaerobic sludge, heavy metals are typically sequestered within extracellular polymeric substances (EPS) and associated with microbial flocs; however, heightened shear forces can compromise floc integrity and induce EPS disintegration, thereby mobilizing previously bound metal ions into the aqueous phase (Li et al., 2017b; Zhang et al., 2016). Such metal release may impair subsequent solid–liquid separation efficiency and influence downstream digestate management. The above results suggest that the TSAD effluent—characterized by comparatively higher heavy metal loads—may present non-negligible environmental risks, particularly upon land application. Consequently, routine monitoring of heavy metal concentrations in system effluents is recommended in operational practice, accompanied by systematic evaluation of potential impacts on soil quality and crop safety.

From a technical and economic standpoint, when the reactor volume is normalized to 1 m^3, the TSAD process demonstrates enhanced processing capacity relative to SSAD due to its integrated strategy of high solid feeding, staged separation and BSR (Table 4). Specifically, at an HRT of 20 days, the TSAD system achieved a daily organic loading rate of 5.30 kg, whereas the SSAD system attained only 1.89 kg/day at an HRT of 30 days, representing a 2.8-fold improvement in processing capacity. This enhancement is primarily attributable to the integrated aerobic hydrolysis stage in TSAD, which facilitates effective pretreatment of feedstock, thereby increasing substrate biodegradability alleviates the hydrolytic bottleneck inherent in conventional single-stage digestion. As a result, the system achieves stable operation at higher OLRs—even under shortened HRTs (Vanegas and Bartlett, 2013).

Table 4
Comprehensive comparison of SSAD and TSAD processes.

	SSAD HRT = 30 d	TSAD HRT = 20 d	HRT = 10 d
Effective volume (L)	600	187.5/62.5/312.5	187.5/62.5/312.5
Feed amount (kg)	20.0	23.4	46.9
Feed TS (%)	6	15	15
Raw material feed amount (kg)	1.9	5.3	11.0
Consumption of clean water (L)	18.1	8.5	17.6
Daily emission volume of biogas residue (Moisture content of 70 %, kg)	3.1	8.1	16.9

Notes: The above data were calculated based on a construction volume of 1 m^3.

Furthermore, under a TS concentration of 15 % in the feed, the TSAD process reduced clean water demand: at an HRT of 20 days, its daily water consumption per unit substrate was merely 8.47 L, corresponding to an 83.3 % reduction compared with the SSAD system (18.11 L/day at an HRT of 30 days). This pronounced reduction in BS emission volume not only lowers downstream dewatering and transport costs but also mitigates associated environmental burdens. Meanwhile, under identical HRT conditions, the TSAD system generated 8.14 kg/day of biogas residue (calculated based on a moisture content of 70 %), approximately 2.6 times the residue output of the SSAD system (3.08 kg/day), indicating that this process is better suited to regions with adequate land availability for residue disposal or well-established downstream residue treatment and valorization infrastructure, thereby facilitating the safe and efficient conversion of digested residue into organic fertilizers or other value-added products.

To further elucidate the technical positioning and novelty of the TSAD process, a systematic performance comparison was conducted against representative AD configurations widely reported in recent literature (Table 5). Specifically, relative to the two-phase hyperthermophilic–mesophilic co-digestion process reported by Chen et al. (2021) (HRT = 30 days; CH_4 yield = 510 mL/g VS), the TSAD process achieves a 33 % reduction in HRT while enabling more precise enrichment of functional microbial communities—facilitated by spatial separation of hydrolysis and acidification zones—and avoids the energy losses from the switching between hyperthermophilic and mesophilic conditions. In comparison with the two-phase system with double thin-slurry reflux described by Janesch et al. (2025) (HRT = 20 days; CH_4 yield = 260 mL/g COD), the TSAD process exhibits a marginally lower CH_4 yield but demonstrates distinct advantages in volumetric organic loading capacity and reduction of digestate volume emissions. Furthermore, many high-yield strategies reported in the literature

Table 5
Comprehensive comparison of the proposed TSAD system with the previous AD process.

Process Configuration	Feedstock	HRT /Pretreatment time	Chem/Bio Additives Needed?	Fermentation temperature	Methane yield	Slurry recirculation?	Reference
TSAD	PM + RS	20 d	None	37℃/37℃/37℃	218 mL/g COD	Liquid slurry recirculation	–
Two-phase Thermophilic-Mesophilic Co-digestion	Swine manure + Rice straw (3:1)	30 d (6 + 24 d)	None	55℃/37℃	510 mL/g VS	Digestate recirculation (ratio 1.0)	(Chen et al., 2021)
Two-phase AD	Food waste	26 d (4 + 22 d)	None	35℃/35℃	>370 mL/g	No recirculation	(Feng et al., 2021)
Single-stage AD + Microaerobic pretreatment	Rice straw	Pre-treatment 24 h + AD 50 d	Microbial reagents (SC)	37℃	311.7 mL/g VS	No recirculation	(Amin et al., 2021)
Two-phase Synergistic Reflux (TPSR)	Corn straw + Pig manure	31 d (1 + 30 d)	None	37℃/37℃	423.65 mL/g VS	Two-phase synergistic recirculation (reflux ratio up to 100 %)	(Yue et al., 2023)
Two-stage + Double thin-slurry recirculation	Maize silage + Bedding straw (50:50)	20 d (10 + 10 d)	None	35-37℃/35-37℃	260 mL/g COD	Thin-slurry recirculation (10 % feed volume)	(Janesch et al., 2025)
Two-stage (HRT/OLR optimized)	Rice straw	38 + 19 d	KOH (pretreatment)	35℃/35℃	~307.4 mL/g VS	No recirculation	(Yang et al., 2024)
Two-stage Thermophilic (carrier materials compared)	Dairy wastewater	Acid: 10 h / Meth: 0.5–9 d	None	55℃/55℃	Up to 305.5 mL/g COD (carbon felt)	No recirculation	(Mikheeva et al., 2024)
Two-stage (batch optimization + semi-continuous)	Filter cake + Biogas effluent	Acid: 3 d / Meth: 20 d	None	37℃/37℃	140.2 mL/g VS	No recirculation	(Wongarmat et al., 2025)
Single-stage AD (CSTR) + Biochar	Food waste	75 h	Biochar	37℃	Not specified (energy yield improved 73.8 %)	No recirculation	(Yuan et al., 2023)

depend either on exogenous fortifiers (Amin et al., 2021; Mikheeva et al., 2024; Yuan et al., 2023), or external acid–base addition for pH stabilization (Feng et al., 2021). By contrast, the TSAD process operates without chemical or biological augmentation, yet achieves a relatively high CH_4 production efficiency, thereby substantially lowering pretreatment costs and operational complexity.

In summary, the TSAD process offers compelling advantages in terms of processing capacity, water conservation, and reduction of BS, making it particularly suitable for centralized agricultural waste treatment facilities with high resource recovery objectives. Its optimal deployment hinges on the rational selection of HRT to mitigate ammonia inhibition risks, coupled with integrated residue disposal and heavy metal management. Future research will focus on validation within an independently replicated pilot-scale reactor system, and further investigate the repeatability under varying feedstock compositions and operating conditions, thereby providing more scientific foundation for the engineering promotion of the TSAD process.

4. Conclusion

The operational performance of TSAD process under varying HRTs were systematically evaluated. Results demonstrated that TSAD enabled directional enrichment of functional microbial communities by physically separating the hydrolysis, acidogenesis and methanogenesis stages, thereby enhancing its resilience to high OLRs. Under an HRT of 10 days, the system achieved a volume biogas production rate of 1665 mL/L/d with a CH_4 content of 61.9 %, indicating excellent short-term biogas production potential. Nevertheless, prolonged operation (120 days) at this HRT induced progressive accumulation of TAN (4855.1 mg/L), causing pronounced inhibition of methanogenic activity. Concurrently, the residual concentrations of heavy metals (Cu and Zn) increased in the liquid phase, thereby elevating environmental risk. In contrast, under an HRT of 20 days, the system achieved stable and efficient biogas production performance over the same 120-day period. By integrating high-solid feedstock, staged separation and BSR, it delivered a processing capacity approximately 2.8-fold higher than the traditional SSAD process, while reducing specific water consumption per unit substrate by 83.3 %, thereby effectively achieving the reduction of BS emission. These findings demonstrate that the TSAD system's long-term operational stability necessitates careful HRT selection coupled with heavy metal immobilization and targeted ammonia nitrogen removal from recirculated slurry, so as to turn potential risks into resource recovery opportunities.

CRediT authorship contribution statement

Kaitao Peng: Writing – original draft, Methodology, Investigation, Conceptualization. **Min Sun:** Investigation, Funding acquisition. **Wei Zeng:** Investigation, Funding acquisition. **Qi Zhang:** Visualization, Methodology. **Yunpu Wang:** Validation, Supervision, Resources, Project administration. **Shuyu Xiang:** Visualization, Methodology. **Yuhuan Liu:** Validation, Supervision, Resources. **Xian Cui:** Writing – review & editing, Visualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2026.135455.

Data availability

No data was used for the research described in the article.

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